Q11 Discussion Questions

1. Kalai and Vempala show calibrated models *must* hallucinate on arbitrary facts, yet post-trained models hallucinate less and are measurably *less* calibrated. Is trading calibration for caution the right deal? What does a user lose when a model's probabilities stop meaning what they say?
2. The module argues hallucination is not lying because there is no intent. Where does that line actually sit? If a model reliably maintains a false claim under clarification ("are you sure?") because refusals are low-probability under its distribution, is that closer to confabulation, stubbornness, or deception — and does the moral category change any engineering decision?
3. Retrieval can inject errors: Mallen et al. found retrieval *hurts* on popular entities the model already knows, and a RAG system inherits its index's failures. Given that, when would you *not* ground a production system — and what audit would you run on the index before trusting it?
4. Gekhman et al. find fine-tuning on new factual knowledge linearly increases hallucination. If SFT teaches the form of answers rather than facts, what should a domain-adaptation team do whose corpus is 90% new knowledge (e.g., internal company facts)? Defend a concrete pipeline.
5. The module's taxonomy separates fabrication, misattribution, and faithful errors. Design an evaluation that reports all three separately for a RAG assistant — what labeled data would you need, and which of TruthfulQA, cloze probes, or HaluEval could you reuse for which slice?